

Vignesh S

Software Engineer

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PROFESSIONAL SUMMARY

Full-stack Software Engineer with nearly 2 years of experience building multi-tenant and customer-facing platforms with React, Next.js, Angular, Node.js, and MongoDB. Built queue-based ingestion that cut database writes by up to 98%, and shipped idempotent online payments on a live platform with 200+ users.

EXPERIENCE

Software Engineer

AAPGS Private Limited

Feb 2025 – Oct 2026

Madurai, Tamil Nadu

- Cut ingestion API response time **from about 900 ms to 462 ms** and reduced **MongoDB writes by up to 98%** across **50+ tenants** by moving database inserts and automation execution off the request path to a **Redis and BullMQ job queue**, batching **50 inserts per write**.
- Shipped a live e-commerce platform (Next.js, Node.js, MongoDB) with **200+ users** and **100+ products**, featuring admin-managed dynamic pages with caching, cash on delivery, and PayU online payments using checkout-scoped idempotency keys to prevent duplicate payment attempts.
- Designed a multi-tenant architecture with JWT authentication, dynamic per-tenant database switching, and an in-memory connection cache with **10-minute TTL eviction**, keeping each tenant's data in its own database.
- Built an interactive workflow automation builder with Foblex Flow where users chain **3 action node types** (HTTP, email, WhatsApp) with pass/fail branching and AND/OR threshold conditions, supporting up to **5 retries** per HTTP request, custom email templates, and a time-series execution history of every run.
- Developed a real-time dashboard system (Angular, Socket.IO, ECharts) with drag-and-drop chart and table widgets and per-dashboard rooms for targeted updates; removed per-emit MongoDB reads by caching dashboard metadata in Redis with NX locks.
- Optimized the **CI/CD pipeline**, reducing deployment time **from 15–20 minutes to about 7 minutes** by eliminating repeated dependency reinstalls, fixing out-of-memory build failures, and lazy loading heavy modules.

Web Developer

E-Careerpluz Private Limited

Dec 2024 – Feb 2025

Madurai, Tamil Nadu

- Built a **40+ page** responsive digital-services website (HTML, CSS, JavaScript, PHP) with contact form handling, email subscription, and automated notifications.

PROJECTS

DeployDock | *Next.js, Node.js, TypeScript, MongoDB, SSH, Docker, SSE* | [GitHub](#)

Open Source

- Built a self-hosted Continuous Integration/Continuous Deployment (CI/CD) platform with SSH-based remote deployments, PM2 process management, and git-based rollback support.
- Automated deployments via GitHub webhook integration, supporting both SSH-based and Docker-based deployment targets while tracking commit SHAs and full deployment history.
- Streamed deploy and PM2 logs in real time via SSE backed by worker threads into a terminal-style viewer with per-line classification and collapsible sections, keeping the pipeline non-blocking under concurrent jobs.
- Built a React Flow visual automation editor for post-deploy actions, sending alerts through **4 providers** (SMTP, ZeptoMail, Twilio, Meta WhatsApp), alongside cron scheduling and per-project pipeline configuration.
- Designed a multi-tenant backend with per-tenant MongoDB databases, role-based access control with custom roles and team invites, and AES-256 encryption of SSH credentials and environment variables written to servers over SSH.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, SQL, Python, C, C++

Frontend: React, Next.js, Angular

Backend: Node.js, Express.js, Fastify, REST APIs, WebSockets, SSE, JWT, BullMQ

Databases: MongoDB, Redis, MySQL

Cloud & DevOps: AWS (S3, EC2, Lambda), GitHub Actions (CI/CD), Docker, PM2, Git, Webhooks

Concepts: Multi-tenancy, RBAC, Message Queues, Caching, Idempotency, Load Testing

EDUCATION

The Madura College

Bachelor of Science in Information Technology, CGPA: 7.61

Madurai, Tamil Nadu

Aug 2021 – May 2024